

1a.) $x^2 - 5x + 6 > 0$ } $y > 0$
 $y = x^2 - 5x + 6$

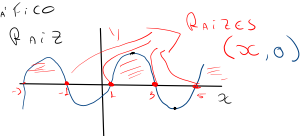
x $y = x^2 - 5x + 6$ y
 Domínio $f(x)$
 Eixo x Imagem Eixo y

Equação

$y = x^2 - 5x + 6$

Gráfico

Cálculo Raiz



$y = x^2 - 5x + 6$

$0 = x^2 - 5x + 6$

$x^2 - 5x + 6 = 0$ } $a = 1$
 $ax^2 + bx + c = 0$ } $b = -5$
 $c = 6$

$\Delta = b^2 - 4ac$

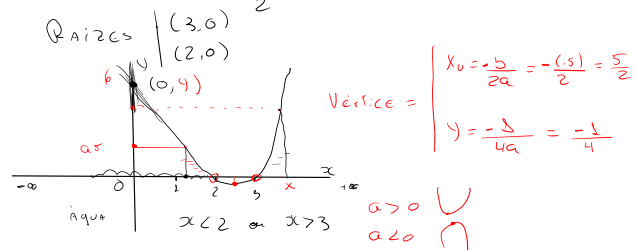
$\Delta = (-5)^2 - 4 \cdot 1 \cdot 6$

$\Delta = 25 - 24$

$\Delta = 1$

$x = \frac{-b \pm \sqrt{\Delta}}{2a}$ $x_1 = \frac{5 + 1}{2} = 3$ (3, 0)

$x = \frac{-(-5) \pm \sqrt{1}}{2}$ $x_2 = \frac{5 - 1}{2} = 2$ (2, 0)



b.) $x^2 - 5x + 6 < 0$ } $y < 0$
 $y = x^2 - 5x + 6$ } $2 < x < 3$

c.) $x^2 - 5x + 6 \leq 0$ } $y \leq 0$
 $y = x^2 - 5x + 6$ } $2 \leq x \leq 3$

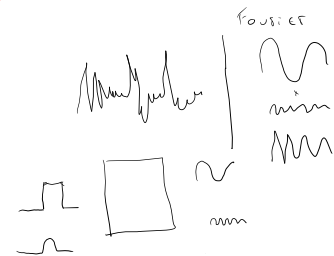
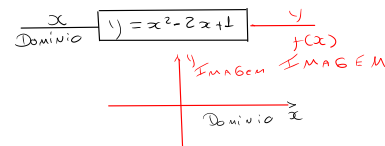
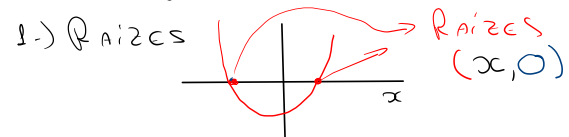


Gráfico Raizes Verice Ponto que o eixo y
 $y = x^2 - 2x + 1$



Gráfico

$$y = x^2 - 2x + 1$$



$$y = x^2 - 2x + 1$$

$$\nabla \quad 0 = x^2 - 2x + 1 \quad \nabla$$

$$x^2 - 2x + 1 = 0$$

$$\begin{cases} a = 1 \\ b = -2 \\ c = 1 \end{cases}$$

$$\Delta = b^2 - 4ac$$

$$\Delta = (-2)^2 - 4 \cdot 1 \cdot 1$$

$$\Delta = 4 - 4 \cdot 1 \cdot 1$$

$$\Delta = 0$$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$x = \frac{-(-2) \pm \sqrt{0}}{2}$$

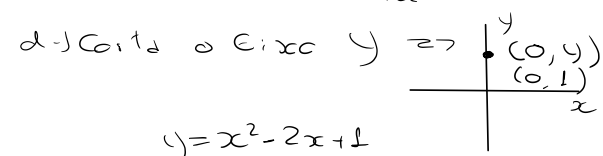
$$x_1 = x_2 = \frac{2}{2} = 1$$

$$x_1 = x_2 = (1, 0)$$

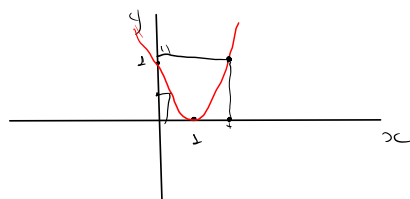
b.) $C_0 \cup C_1 \cup \dots \cup C_n \in \mathcal{C}$ $\alpha \in \mathcal{C}$ $\cup$

c.) $\forall x \in \mathbb{R} : c \in \mathbb{R}$

$$\begin{cases} x_0 = \frac{-b}{2a} = \frac{-(-2)}{2} = 1 \\ y_0 = \frac{-\Delta}{4a} = 0 \end{cases} \quad (1, 0)$$



$$\forall x = 0 \Rightarrow y = 1$$



d.) $x^2 - 2x + 1 > 0 \Rightarrow y > 0$

$$y = x^2 - 2x + 1$$

$$V = \mathbb{R} - \{1\}$$

e.) $x^2 - 2x + 1 < 0$

$$V = \emptyset$$

f.) $x^2 - 2x + 1 \geq 0$

$$V = \mathbb{R}$$

h.) $x^2 + x + 1 > 0$

$\frac{x}{\text{Dominio}} \quad y = x^2 + x + 1 \quad \text{Imagen } f(x)$

$$h.) \left. \begin{aligned} & x^2 + x + 1 > 0 \\ & y = x^2 + x + 1 \end{aligned} \right\} y > 0$$

$$V = \mathbb{R}$$

$$y = x^2 + x + 1$$

$$\text{Raizes} = (x, 0)$$

$$0 = x^2 + x + 1$$

$$x^2 + x + 1 = 0$$

$$\Delta = -3$$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

Não existe
raízes reais

$$x = \frac{-1 \pm \sqrt{-3}}{2}$$

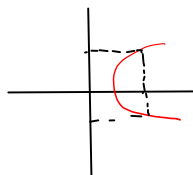
$$y = \sqrt{4-x^2}$$

$$\begin{array}{c} x \\ \text{Domínio} \end{array} \boxed{y = \sqrt{4-x^2}} \begin{array}{c} y \\ f(x) \\ \text{Imagem} \end{array}$$

$$\sqrt{A} \Rightarrow A \geq 0$$

$$4-x^2 \geq 0$$

$$\text{Se } (4-x^2) \geq 0 \text{ então } (4-x)^{\frac{1}{2}}$$



$$\begin{array}{c} x \\ \text{Domínio} \end{array} \boxed{y = 2} \begin{array}{c} y \\ f(x) \end{array}$$

